CE/CS 4390.001/003/004/5390.003 – Computer Networks
Spring 2026
Project
Date: February 26, 2026
Due: April 16, 2026 @ 11:59 PM

In this project, you are required to implement a network application using JAVA. It should use key concepts of intercommunication between programs running on different computers in the network by using Java Sockets. **The application consists of a centralized Math server and two or more clients. The server will provide basic math calculation services to the client(s).**

You need to write a server application with the following requirements:

1. The server keeps track of all users – track who, when, and how long the user is attached to the server.
2. The server should wait for the client's request. Upon attachment, the server should log the client's details.
3. The server can have connections simultaneously with multiple clients.
4. The server should be able to accept the string request for basic math calculations and show who has sent what request.
5. The server should respond to clients in order of the requests it gets from different clients.
6. The server should close the connection with the client upon request from the client and log it.

You need to write a client application with the following requirements:

1. The client gives its name during initial attachment to the server and waits till it gets an acknowledgement from the server for a successful connection.
2. After a successful connection, the client can send basic math calculation requests (at least 3) to the server at random times.
3. Another client can join during this time and start sending its requests.
4. The client should send a close connection request to the server, and the application terminates.

Design your protocol:

1. Message format for sending and receiving math calculations.
2. Message format for joining and terminating the connection.
3. Format for keeping logs of clients' activities on the server side.

For CS 5390 students only:

1. Add the functionality of group chat between clients. Clients should be able to chat with other clients regarding equations.

Notes:

- You do not need to design a GUI for the server and client applications.
- Specify your assumptions clearly.
- The requirements mentioned above are for open interpretation, but with acceptable reasoning. You should reach out to the TA or me if you need more clarification on the requirements.

- This project will count for 100 points.
  - 30 points: Design and implementation of the Server application.
  - 30 points: Design and implementation of the Client application.
  - 20 points: Design and use of communication protocol/format.
  - 20 points: Documentation and quality of code.

Submission requirements:
At end of the semester, you must submit a zip file containing

1. Project report (Word/PDF) format covering
   a. Names and NetIDs for the group members
   b. Project partners' contributions
   c. Protocol design
   d. The programming environment you used.
   e. How to compile and execute your programs.
   f. Parameters needed during execution (i.e., IP, port, or name)
   g. Good use of comments throughout your files and code
   h. If your application is not complete, specify what works and what doesn't.
   i. Challenges faced.
   j. What have you learned doing the project?
   k. The output screenshots of the application
2. All codes are needed to run your application, and a Make file
3. A design document describing your implementation of the network application.

Please reach out to the TA or me in case of any questions or concerns.